Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

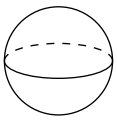
Section 1: Properties of Cylinders and Spheres

Let's look closely at two three-dimensional shapes you meet all the time: the cylinder and the sphere. A cylinder is the shape of a tin or a drum — straight sides with a round top and a round base. A sphere is the shape of a ball or an orange — perfectly round.

Look at the cylinder in fig-01. It has a flat circular top and a flat circular base, and between them one face that curves all the way around the sides. So a cylinder has two circular faces and one curved face. Count them together and you find it has three faces altogether. Run your finger along the rim of the top circle: that is one curved edge, and the rim of the base circle is another. So it has two curved edges. Notice, too, that there is no sharp corner anywhere on a cylinder — no point where three or more edges meet. That means a cylinder has no vertex at all.

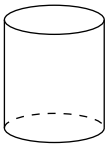
Now look at the sphere in fig-02. A sphere is a single curved surface, with nothing flat on it. There are no edges and no vertices anywhere on it. Finally, turn the sphere any way you like: it looks the same from every direction, so we say it is symmetrical from all directions.

Visual



A sphere drawn as a circle with an ellipse across its middle to show the curved surface.

Visual



A cylinder drawn upright: an ellipse for the top face, two vertical sides falling from its ends, and a curve closing the base.

Let's identify the properties of a cylinder together, one by one.

Look again at the cylinder in fig-01. Ask yourself four questions.

First, what faces does it have? You can see a flat circle at the top and a flat circle at the bottom, and the curved side running between them. That is two circular faces and one curved face — three faces in total.

Second, does it have any vertices? A vertex is a sharp corner where edges meet. Run your eye around the whole cylinder: there is no sharp corner anywhere. So it has no vertex.

Third, does it have any edges? An edge is a line where two faces meet. Where the curved face meets the top circle, there is a curved rim — that is one curved edge. Where it meets the base circle, there is another. So it has two curved edges.

Now do the same for the sphere in fig-02. Ask the same questions. It has only one face, and that face is curved — there is no flat circle anywhere. With no flat face meeting a curved one, there are no edges. And with no edges, there is no sharp corner, so there is no vertex either. Finally, spin the sphere: it looks the same from every direction, which is why we say it is symmetrical from all directions.

So you can list the properties of each shape in the same way: count the faces, count the edges, count the vertices, and note any special feature such as the cylinder's curved faces and edges, or the sphere's symmetry.

Practice

1. How many circular faces does a solid cylinder have?
2. How many faces does a cylinder have altogether?

3. State how many edges and how many vertices a sphere has.
4. A pupil says a cylinder has one curved face and two curved edges. Is the pupil right about the edges?

Section 2: Volume of a Cube, a Cuboid and a Prism

Volume tells us how much space a three-dimensional shape takes up, and it is measured in cubic units like cm^3, m^3. For a cuboid, think of it as a box that is made of layers of cubes stacked up. If you know the area of the base (how many cubes cover the floor of the box) and how many layers high it is, you can find the volume.

Volume of a cuboid = area of base × height = length × breadth × height, often written simply as $\mathbf{V = l \times b \times h}$ or $\mathbf{V = lbh}$.

Volume of a cube is the same idea, but a cube has every edge the same length. So its volume is length × length × length, written as $\mathbf{V = l^3}$.

A prism is a solid whose two ends are identical. For any prism, you can again find the volume by multiplying the area of one end (the base) by the length it runs back: **Volume of a prism = area of base × height**. For a triangular prism, the base is a triangle, so you first find the area of that triangle using area = ½ × base × height of the triangle, then multiply by how far the prism runs.

Example 1. Calculate the volume of a cuboid of length 3 cm, breadth 2 cm and height 8 cm.

Volume of cuboid = l × b × h.

length = 3 cm, breadth = 2 cm, height = 8 cm.

Volume = 3 cm × 2 cm × 8 cm = 48 cm^3.

Example 2. The volume of a cuboid is 24 cm^3. Its base is 6 cm by 2 cm. Find its height.

Volume of cuboid = l × b × h.

length = 6 cm, breadth = 2 cm, height = h.

Volume = 24 cm^3.

So 24 cm^3 = 6 cm × 2 cm × h.

24 cm^3 = 12 cm^2 × h.

Divide both sides by 12 cm^2:

24 cm^3 ÷ 12 cm^2 = h, so h = 2 cm.

Example 3. Calculate the volume of a cube of side 3 cm.

Volume of cube = l^3.

V = $(3\text{ cm})^3$ = 3 cm × 3 cm × 3 cm = 27 cm^3.

Example 4. fig-03 shows a triangular prism lying on its rectangular face. The triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m. Find its volume.

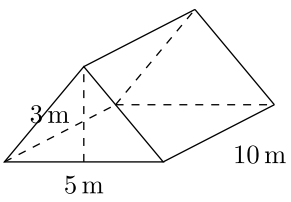
First find the area of the triangular end (the base of the prism):

Area of triangle = ½ × base × height = ½ × 5 m × 3 m = 7.5 m^2.

Now multiply by the length the prism runs:

Volume of prism = area of base × height = 7.5 m^2 × 10 m = 75 m^3.

Visual



A triangular prism lying on its rectangular face. The triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

Practice

1. Calculate the volume of a cuboid 8 cm long, 4 cm wide and 3 cm high.
2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
3. What is the volume of a cube measuring 4 cm × 4 cm × 4 cm?
4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 m³ of water. How deep is the water in the tank?

Check your work

Attempt every question before you read this page.

Properties of Cylinders and Spheres

1. Two — one circular top face and one circular base.

   A solid cylinder has a flat circular top and a flat circular base, so it has two circular faces.
2. Three — two circular faces and one curved face.

   Count the flat circular top, the flat circular base, and the curved face that runs between them: that gives three faces altogether.
3. A sphere has no edges and no vertices.

   A sphere is one single curved surface with nothing flat on it. Since edges form where faces meet and vertices form where edges meet, a sphere has neither.
4. Yes — the two curved edges are the rim of the top circle and the rim of the base circle.

   Each rim is a line where the curved face meets a circular face. There is one such rim at the top and one at the base, so the cylinder has two curved edges.

Volume of a Cube, a Cuboid and a Prism

1. 96 cm³

   $V = l \times b \times h = 8\text{ cm} \times 4\text{ cm} \times 3\text{ cm} = 96\text{ cm}^3$.
2. 30 m³

   $V = 5\text{ m} \times 2\text{ m} \times 3\text{ m} = 30\text{ m}^3$.
3. 64 cm³

   $V = l^3 = (4\text{ cm})^3 = 4\text{ cm} \times 4\text{ cm} \times 4\text{ cm} = 64\text{ cm}^3$.
4. 3 m

   Volume = l × b × h, so $36\text{ m}^3 = 6\text{ m} \times 2\text{ m} \times h = 12\text{ m}^2 \times h$. Divide both sides by 12 m²: $h = 36\text{ m}^3 \div 12\text{ m}^2 = 3\text{ m}$.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-a86f099b	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____